

Interactions of pay-on-delivery service and distribution contracts on e-commerce platforms

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Abstract Pay-on-delivery allows consumers to postpone payment until product receipt and can mitigate the online channel’s disadvantage caused by delivery delay. This study presents the first analytical research on when the pay-on-delivery service should be offered and how it impacts the manufacturer’s choice of distribution contracts in a dual-channel supply chain. We develop game-theoretic models to characterize the equilibrium adoption of pay-on-delivery and the resulting pricing, demand, profit, and welfare outcomes. The results show that pay-on-delivery can mitigate, and under some conditions even reverse, the online channel’s disadvantage caused by delivery delay. Its adoption is governed by the trade-off between consumers’ valuation discount associated with delayed receipt and firms’ opportunity cost associated with deferred payment. We further show that pay-on-delivery reshapes the manufacturer’s contract choice by changing the relative attractiveness of the online channel for consumers and delaying cash recovery of channel members. Additionally, although pay-on-delivery may reduce consumer surplus, it improves social welfare.

Keywords: Pay-on-delivery service; Distribution contracts; Dual-channel; Game theory

1. Introduction

E-commerce has grown rapidly over the past decades and become a central component of modern retailing. Global e-tailer sales surpassed an impressive \$6.42 trillion in 2025 and are projected to surge to \$7.89 trillion by 2028 (Lin, 2025). Against this backdrop, many manufacturers no longer rely on a single sales channel. Instead, they cooperate with e-commerce platforms to complement their traditional offline channel and broaden consumer reach. This pattern is clearly observed in practice. For example, IKEA launched a digital store on JD.com in 2025 while continuing to operate 40 physical stores in China (Reuters, 2025). Consistent with this development, recent operations research has increasingly examined dual-channel structures in which a manufacturer sells offline while collaborating with an e-commerce platform online, and in which the online selling format may take the form of agency selling or reselling (Li et al., 2024; Liu et al., 2024).

Alongside the rapid expansion of platform-based online selling, payment design has emerged as an increasingly important operational lever. One prominent innovation is the pay-on-delivery service, which allows consumers to defer payment until product receipt rather than paying at the time of ordering (Liu et al., 2019; J. Zhang et al., 2021). Although digital payment tools have become widespread, pay-on-delivery remains actively used in major e-commerce markets. Amazon continues to offer cash-on-delivery or pay-on-delivery options for eligible orders in markets such as India and the United Arab Emirates, and Noon continues to list cash on delivery as a payment option in markets such as Saudi Arabia and Egypt (Amazon, 2026a, 2026b; Noon, 2026). These developments suggest that payment timing is not merely a checkout detail. For manufacturers that maintain offline channels while relying on platforms to expand online sales, it is a strategically relevant channel design choice that may materially influence the attractiveness and performance of the online channel.

Yet, the adoption of pay-on-delivery is far from straightforward. For a manufacturer that already operates an offline channel, the online channel faces an inherent disadvantage. Consumers typically place orders and make payments before receiving the product, whereas offline purchasing offers immediate acquisition. By postponing payment until delivery, pay-on-delivery can mitigate this disadvantage and make the online channel more attractive to consumers, especially when delivery delay weakens the appeal of online purchasing (Liu et al., 2019; J. Zhang et al., 2021). However, this same timing change also delays revenue realization and cash recovery for the selling side. Once payment is shifted from the ordering stage to the delivery stage, the party providing the online channel must bear the opportunity cost of deferred cash

inflows, such as delayed capital turnover and increased financing pressure. Pay-on-delivery therefore creates a fundamental operational trade-off. It may stimulate online demand by improving consumers' payment timing, while at the same time reducing the profitability of the channel member that bears the delayed cash flow. In a dual-channel setting, this trade-off becomes particularly consequential because any improvement in the online channel may also reshape competition with the manufacturer's offline channel.

This trade-off becomes even more intricate once the manufacturer's online channel can be organized under different distribution contracts. A growing stream of platform and supply chain research shows that, when selling through an e-commerce platform, manufacturers commonly face a choice between a wholesale (or reselling) format and an agency selling format, and that this choice materially affects pricing authority, revenue allocation, and channel performance (Abhishek et al., 2016; Li et al., 2024; Liu et al., 2024; Matsui, 2024). Under a wholesale contract, the platform purchases from the manufacturer and resells to consumers. The platform therefore controls the online retail interface, independently decides whether to offer pay-on-delivery services, and directly bears the consequence of delayed consumer payment. Under an agency contract, by contrast, the manufacturer sells directly to consumers on the platform and pays the platform a commission fee. In this setting, the implementation of pay-on-delivery requires the consent of both the manufacturer and the platform. Once adopted, the service postpones consumer payment until delivery and therefore delays revenue realization for both parties. Consequently, the implications of pay-on-delivery vary across distribution contracts. Beyond its direct effect on online demand, it changes who holds pricing control, who bears the opportunity cost from delayed payment, and how the resulting online competition affects the offline channel. This interplay highlights the need to jointly consider pay-on-delivery adoption and distribution contract choice in a dual-channel supply chain.

Motivated by these observations, we examine three interrelated questions in a dual-channel supply chain where a manufacturer sells offline and collaborates with an e-commerce platform online. First, under what conditions should the pay-on-delivery service be adopted under different distribution contracts? Second, how does the pay-on-delivery service influence optimal profits, consumer surplus, and social welfare? Third, once pay-on-delivery becomes feasible, how does it affect the selection of distribution contracts? To address these questions, we develop game-theoretic models to characterize the equilibrium adoption of pay-on-delivery and to examine how payment timing interacts with contractual structure in shaping channel outcomes. We also extend the main model to assess the robustness of the main results and to derive additional managerial insights.

This paper makes three contributions. First, it identifies the economic conditions under which pay-on-delivery becomes viable in a dual-channel supply chain. In particular, the adoption of pay-on-delivery is jointly governed by consumers' valuation discount associated with delayed receipt and firms' opportunity cost associated with deferred payment. Second, the paper shows that pay-on-delivery can materially reshape the manufacturer's preferred mode of online cooperation. Specifically, pay-on-delivery changes the relative attractiveness of the online channel for consumers and delays cash recovery of channel members, thereby altering the relative profitability of the two selling formats for the manufacturer. Third, the paper demonstrates that pay-on-delivery has broader channel and welfare implications beyond payment convenience alone. It can mitigate price competition, alter offline-online cannibalization, and generate nontrivial implications for consumer surplus and social welfare. We further extend the analysis to incorporate product returns and show that the strategic appeal of pay-on-delivery may become stronger when return-related considerations are taken into account.

The paper is organized as follows. Section 2 reviews the related literature. Section 3 outlines the model setting, and the results are analyzed in Section 4. In Section 5, we extend our models and we provide the conclusion in Section 6.

2. Literature review

This study contributes to extending current research on the dual-channel supply chain by investigating the interplay between the pay-on-delivery service and the selection of distribution contracts. As such, our research is connected to two literature streams: the payment scheme in transactions and the selection of distribution contracts.

2.1. *The payment scheme in transactions*

In this field, some scholars focus on the payment scheme between suppliers and retailers (Bhattacharya et al., 2019; Cao et al., 2022; Mandal et al., 2024; Yan & He, 2020; Wang et al., 2025), while others concentrate on that for consumers. For the sake of brevity, we focus only on the latter one which is most related to our research. With the continued expansion of e-commerce, online retailers offer various payment methods to consumers to enhance their shopping experience, primarily divided into pay-to-order and pay-on-delivery (See-To et al., 2014; Xu et al., 2017). Under traditional pay-to-order methods, payment must be made when consumers order the products, while the pay-on-delivery service allows them to pay upon receipt. In addition to pay-on-delivery, e-commerce platforms may introduce credit payments to bolster market competitiveness. Although pay-on-delivery and credit payment share similar characteristics of delayed payments, they possess fundamental differences. Credit payment installment is designed to accommodate consumers with insufficient budgets, who are required to repay the bill on a predetermined time. The firms are primarily concerned with the risk of delinquent repayment (Q. Li et al., 2023; Wei et al., 2023; Xie et al., 2021; Zha et al., 2022). In contrast, pay-on-delivery is designed to alleviate logistics-related concerns by allowing payment upon receipt of the product. Furthermore, this service enables consumers to directly return unsuitable products upon delivery. Firms are more focused on their cash flow and potential product returns (Liu et al., 2019; Xu et al., 2017; J. Zhang et al., 2021).

Previous studies typically concentrate on what drives consumers to use the pay-on-delivery service, such as building trust (Tandon et al., 2016) and increasing purchasing intentions (Thakur & Srivastava, 2015). The influence of payment services, such as pay-on-delivery, on operations management has drawn more academic attention in recent years (Chen et al., 2019; Feng et al., 2022; Huang et al., 2022; Liu et al., 2019; Y. Shi et al., 2022; Xu et al., 2017; Xu et al., 2025; J. Zhang et al., 2021). Chen et al. (2019) analyze the effects of order-based payment, delayed payment, and prepayment on inventory decisions, providing references for supply contract design and chain financing. Liu et al. (2019) consider the decision-making of pay-on-delivery service lead times and their impact on the manufacturer's encroachment and find that the encroachment can be more aggressive under pay-on-delivery. Xu et al. (2017) and X. Zhang et al. (2021), both considering the convenience of product returns with pay-on-delivery services, study optimal pricing decisions and joint pricing-inventory decisions respectively. Feng et al. (2022) explore the impact of payment methods on retailers' ordering strategies, finding that delayed payment acts as a method to increase order quantities. Xu et al. (2025) study the pricing decisions for online sellers under pay-to-order and pay-on-delivery schemes, showing that pay-on-delivery is preferred when the service cost or return handling cost is sufficiently small.

This study is closely related to Liu et al. (2019), who focus on a manufacturer with a wholesale offline channel and an online channel that has decided to provide the pay-on-delivery service. However, our research concentrates on a market where a manufacturer with a direct offline channel is considering cooperation with an e-commerce platform. We complement their work by examining the conditions under which adopting pay-on-delivery services is beneficial and how it influences the manufacturer's choice of distribution contract. Meanwhile, we allow the manufacturer to refuse the pay-on-delivery service, given that the successful implementation of the pay-on-delivery service under the agency contract relies on the manufacturer's willingness to accept this service. In the main model, we isolate the intertemporal mechanism of pay-on-delivery, namely delayed receipt and delayed payment, while the inspection/return-related benefit is incorporated in Section 5 as an extension.

2.2. The selection of distribution contracts

This paper also complements the burgeoning research stream on the selection of distribution contracts, especially between the generally adopted wholesale contract (e.g., BestBuy.com) and agency contract (e.g., Zappos.com). Prior research has studied how the factors impact the choice of online distribution contract, including product reviews (Kwark et al., 2017), service effort level (Ha et al., 2022; Xu et al., 2024), quality differentiation (Yenipazarli, 2021; Zhang et al., 2019), product returns (Hao & Yang, 2023; Huang et al., 2024), information sharing (Gong et al., 2024; Lin et al., 2024), digital piracy (Mandal & Roy, 2025), fulfillment logistics (Hu & Fu, 2025), and store brand competition (Ke et al., 2024). For a comprehensive review, see Tan et al. (2024). Differently, we investigate how the payment scheme influences the distribution contract, which has not been studied to our knowledge.

Additionally, some scholars concentrate on different supply chain structures, depending on the factors they are concerned about. For example, Abhishek et al. (2016) hypothetically investigate the scenario in which two downstream e-tailers and one upstream supplier should employ agency selling rather than conventional reselling. Tian et al. (2018) and Zennyo (2020) examine two competing suppliers and a platform, investigating how upstream competition between suppliers affects optimal mode selection. Studies concerning a manufacturer utilizing a dual-channel supply chain, including a direct channel and an e-commerce platform, have also emerged (Qin et al., 2020; G. Shi et al., 2022; Zhang & Zhang, 2020; X. Zhang et al., 2021), which is most related to our research.

Most recently, many platforms operate in hybrid mode and the manufacturers can decide how to join the platform. For example, manufacturers can sell products to Amazon, which then sells these products to consumers (i.e., wholesale contract); additionally, selling their products directly to consumers on Amazon Marketplace (i.e., agency contract) is also optional (Zennyo, 2020). Therefore, many studies have emerged by investigating the manufacturer's selection of distribution contracts under different contexts (Chen et al., 2021; Gong et al., 2024; Hu & Fu, 2025; Ke et al., 2024; Lin et al., 2024; Mandal & Roy, 2025; Matsui, 2024; Wei et al., 2020; Zhang et al., 2022; Zhen et al., 2022). For example, Matsui (2024) investigates the choice of distribution contracts of competing manufacturers. Mandal and Roy (2025) examine how digital piracy shapes the preference between agency and wholesale contracts. Gong et al. (2024) study how platform information sharing interacts with the selling format decision when the manufacturer encroaches. We contribute to this stream by examining a dual-channel supply chain that comprises an e-commerce platform and a manufacturer distributing products through both offline and online channels, either under a wholesale contract or agency contract. We analyze how the adoption of pay-on-delivery service impacts the selection of distribution contracts. To our knowledge, the selection of the distribution contracts of the online channel considering payment schemes has not been examined.

Table 1 summarizes the existing studies to show the contribution of our research.

Table 1 Comparison of this research with previous studies

	Pay-on-delivery service	Payment delay	Product returns	Distribution contracts	Dual- channel
J. Zhang et al. (2021)	√		√		
Tandon et al. (2016)	√	√			
Xu et al. (2025)	√	√			
Liu et al. (2019)	√	√			√
Wang et al. (2025)		√			
Mandal et al. (2024)		√			
Tan et al. (2024)				√	√
Matsui (2024)				√	√
Huang et al (2024)			√	√	√
Hao & Yang (2023)			√	√	√
Gong et al. (2024)				√	√
Hu & Fu (2025)				√	
Ke et al. (2024)				√	
Our Paper	√	√	√	√	√

3. Model

We consider a supply chain composed of a continuum of consumers, an upstream manufacturer, and an e-commerce platform. The manufacturer produces a single product, sells it through its offline channel, and cooperates with the platform to open an online channel. Depending on the distribution contract, the platform acts either as a reseller or as an intermediary.

3.1. Channel configuration

The manufacturer operating an offline channel considers partnering with an e-commerce platform to establish an online channel. He can choose from two distribution contracts: (i) Wholesale contract (denoted by W): the manufacturer sells his products to the platform at a wholesale price w , and also sells his product in the offline channel at a retail price p_f^W , in which the subscript f denotes the offline channel; the products are resold to consumers by the platform at the online retail price p_n^W , in which the subscript n denotes the online channel; (ii) Agency contract (denoted by A): the manufacturer sells directly to consumers on the platform but must pay the platform β ($\beta \in (0,1)$) portion of online sales revenue as a commission fee. In this case, the manufacturer simultaneously determines its offline and online retail price, denoted by p_f^A and p_n^A , respectively.

Under the wholesale contract, the online transaction is purely conducted by the platform. The platform therefore has the authority to decide whether to offer the pay-on-delivery service. If the service is provided, the platform bears the cash-flow opportunity cost caused by deferred consumer payment. Following Kouvelis and Zhao (2012), we use γ ($\gamma \in (0,1)$) to denote the revenue discount factor associated with this cash-flow opportunity cost.

Under the agency contract, the online transactions are conducted by the manufacturer through the platform. Accordingly, consumers can use pay-on-delivery only if the platform provides the service and the manufacturer agrees to adopt it. Once the service is adopted, consumer payment is postponed until delivery, which delays revenue realization for both the manufacturer and the platform. In addition, to use the pay-on-delivery service, the manufacturer is required to pay the platform α ($\alpha \in (0,1)$) portion of the online revenue as a service fee.

3.2. Consumer utility and demand specification

Assume that the consumers are heterogeneous in their valuation v of the product, with $v \sim U[0,1]$. Each consumer chooses between the offline channel and the online channel. If the consumer chooses the offline channel, the utility is $u_f = v - p_f - l$, where p_f denotes the offline retail price, and l ($l > 0$) captures the relative shopping-cost disadvantage of the offline channel. This term reflects costs such as travel and in-store shopping effort. We normalize the corresponding online shopping cost to zero, consistent with the idea that searching and browsing online is typically less costly than visiting a physical store (Hao and Kumar, 2023; Tang et al., 2023).

If the consumer purchases online, he will receive the product at a future time. We use the parameter θ ($\theta \in (0,1)$) to capture the valuation discount caused by receipt delay. A lower θ corresponds to a longer waiting time and therefore a greater discount in consumers' valuation of the product. Therefore, the difference between the offline and online channels is jointly captured by l and θ , whereas θ itself only reflects the valuation discount caused by delayed receipt. Under the traditional pay-to-order scheme, the consumer pays at the time of ordering. The utility from buying online is therefore $u_n^N = \theta v - p_n$. When pay-on-delivery is available, the consumer can pay for the product upon receipt and benefit from the delayed payment, as the actual price of the purchased product is also discounted. Following Liu et al. (2019), the discounted future utility with pay-on-delivery service is $u_n^Y = \theta(v - p_n)$. This specification captures the core trade-off studied in the paper. Delayed receipt lowers the effective valuation of the product, whereas delayed payment increases the attractiveness of the online channel by reducing the burden of paying in advance. The main model focuses on this timing mechanism. Section 5 extends the analysis to incorporate the inspection benefit of pay-on-delivery and its implications for product returns, and also investigates the scenario when the discount factors of product valuation and payment are different.

We denote D_f and D_n as the demand of the offline and online channels, respectively. Based on the utility function, we can derive the demand functions as follows:

$$D_f = \begin{cases} 1 - \frac{p_f - p_n + l}{1 - \theta}, & \text{without the pay-on-delivery service} \\ 1 - \frac{p_f - \theta p_n + l}{1 - \theta}, & \text{with the pay-on-delivery service} \end{cases}$$

$$D_n = \begin{cases} \frac{p_f - p_n + l}{1 - \theta} - \frac{p_n}{\theta}, & \text{without the pay-on-delivery service} \\ \frac{p_f - \theta p_n + l}{1 - \theta} - p_n, & \text{with the pay-on-delivery service} \end{cases}$$

3.3. Game sequence and profit functions

Note that decision rights over pay-on-delivery and pricing depend on the distribution contract chosen by the manufacturer. Thus, the sequence of decisions differs across scenarios. There are four possible subgames listed below.

1) *WN*: wholesale contract without pay-on-delivery. In Stage 1, the manufacturer chooses the wholesale contract. In Stage 2, the platform does not offer pay-on-delivery. The profit functions of the manufacturer and the platform are $\pi_m^{WN} = p_f^{WN} D_f^{WN} + w^{WN} D_n^{WN}$ and $\pi_p^{WN} = (p_n^{WN} - w^{WN}) D_n^{WN}$, respectively.

2) *WY*: wholesale contract with pay-on-delivery. In Stage 1, the manufacturer chooses the wholesale contract. In Stage 2, the platform offers pay-on-delivery. The corresponding profit functions are $\pi_m^{WY} = p_f^{WY} D_f^{WY} + w^{WY} D_n^{WY}$ and $\pi_p^{WY} = (\gamma p_n^{WY} - w^{WY}) D_n^{WY}$.

3) *AN*: agency contract without pay-on-delivery. In Stage 1, the manufacturer chooses the agency contract. The platform does not offer pay-on-delivery in Stage 2 or the manufacturer declines to adopt it in Stage 3. Profits are given by $\pi_m^{AN} = p_f^{AN} D_f^{AN} + (1 - \beta) p_n^{AN} D_n^{AN}$ and $\pi_p^{AN} = \beta p_n^{AN} D_n^{AN}$.

4) *AY*: agency contract with pay-on-delivery. In Stage 1, the manufacturer chooses the agency contract. In Stage 2, the platform offers pay-on-delivery, and in Stage 3 the manufacturer adopts it. The profit functions are $\pi_m^{AY} = p_f^{AY} D_f^{AY} + \gamma(1 - \alpha - \beta) p_n^{AY} D_n^{AY}$ and $\pi_p^{AY} = \gamma(\alpha + \beta) p_n^{AY} D_n^{AY}$.

Fig.1. shows the channel configurations under four scenarios.

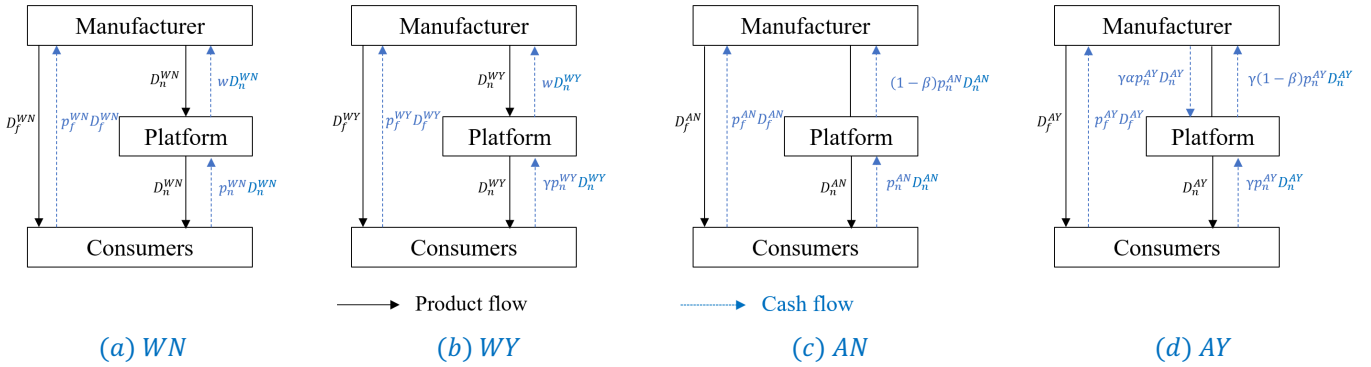


Fig. 1. Illustration of channel configurations under four scenarios.

The overall game sequence is illustrated in Fig. 2. In Stage 1 (S1), the manufacturer selects a distribution contract to collaborate with the e-commerce platform, given that numerous platforms operate different modes that permit the manufacturer to choose his preferred distribution contract. In Stage 2 (S2), the platform decides whether to offer a pay-on-delivery service. In Stage 3 (S3), if an agency contract is selected and the platform provides the pay-on-delivery service, the manufacturer must determine whether to adopt this service. In Stage 4 (S4), under a wholesale contract, the manufacturer decides both the wholesale price and the offline retail price. Finally, in Stage 5 (S5), if a wholesale contract is chosen, the platform sets the online retail price; if an agency contract is selected, the manufacturer determines both the online and offline retail prices.

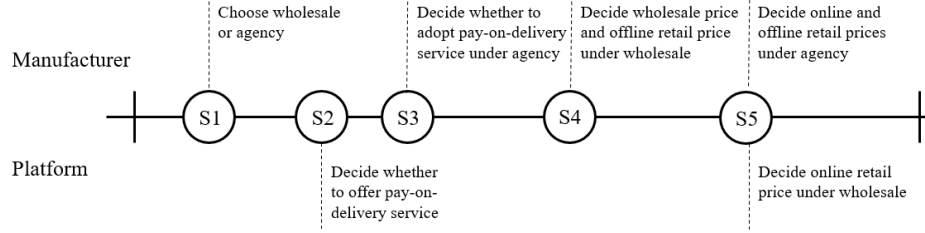


Fig. 2. Game sequence of the model.

We summarize all notations in Table 2. The subscripts denote different channels. The online channel is denoted by n and the offline channel is denoted by f . The superscripts indicate different scenarios, including WN , WY , AN , and AY .

Table 2 Model notations

Notations	Definitions
v	Consumers' valuation of the product, $v \sim U[0,1]$
θ	The discount factor due to receipt delay
γ	The revenue discount factor due to cash-flow opportunity cost with pay-on-delivery service
l	The cost difference between the offline and online channels
α	The commission rate of the pay-on-delivery service under AY scenario
β	The commission rate of the agency contract under AN and AY scenarios
δ	The probability that the product of the online channel fits the consumers' needs
m	Consumers' cost to return products
c	The hassle cost for the seller to handle the return process
w^k	Wholesale price in scenario k , $k \in \{WN, WY\}$
p_i^k	Retail price of channel i in scenario k , $i \in \{n, f\}$, $k \in \{WN, WY, AN, AY\}$
D_i^k	Demand of channel i in scenario k , $i \in \{n, f\}$, $k \in \{WN, WY, AN, AY\}$
π_m^k	Profit of the manufacturer in scenario k , $k \in \{WN, WY, AN, AY\}$
π_p^k	Profit of the platform in scenario k , $k \in \{WN, WY, AN, AY\}$
CS^k	Consumer surplus in scenario k , $k \in \{WN, WY, AN, AY\}$
SW^k	Social welfare in scenario k , $k \in \{WN, WY, AN, AY\}$

4. Analysis

Using backward induction, we begin by examining the four subgames in stage 4, and then characterize the adoption of pay-on-delivery service under different distribution contracts in stage 2 and stage 3. Subsequently, we analyze the manufacturer's choice of distribution contract in stage 1. Finally, we examine how the adoption of pay-on-delivery service impacts the choice of distribution contract.

4.1. Adoption of pay-on-delivery service under the wholesale contract

In this subsection, the platform's decision to provide a pay-on-delivery service under the wholesale contract in stage 2 is examined. To this end, we first determine the equilibrium solutions in both WN and WY scenarios. We then derive the conditions for offering the pay-on-delivery service. Finally, the equilibrium solutions in the two scenarios are compared to illustrate the impact of pay-on-delivery service implementation. The following lemma summarizes equilibrium solutions in both WN and WY scenarios.

Lemma 1. *In WN scenario, the equilibrium solutions are as follows:*

$$\begin{aligned}
(i) \quad & w^{WN*} = \frac{\theta}{2}, p_n^{WN*} = \frac{(2+l)\theta}{4}, p_f^{WN*} = \frac{1-l}{2}; \\
(ii) \quad & D_n^{WN*} = \frac{l}{4-4\theta}, D_f^{WN*} = \frac{2-l(2-\theta)-2\theta}{4-4\theta}; \\
(iii) \quad & \pi_m^{WN*} = \frac{1}{8} \left(2 + l \left(l + \frac{l}{1-\theta} - 4 \right) \right), \pi_p^{WN*} = \frac{l^2\theta}{16(1-\theta)}.
\end{aligned}$$

In WY scenario, the equilibrium solutions are as follows:

$$\begin{aligned}
(i) \quad & w^{WY*} = \frac{\gamma(2(1+l)\gamma-(2+l)\gamma\theta+\theta(2(1-l-\theta)+l\theta))}{(8-\gamma)\gamma-6\gamma\theta-\theta^2}, \quad p_n^{WY*} = \frac{\theta(1-l-\theta)+\gamma(3(1+l-\theta)-2l\theta)}{(8-\gamma)\gamma-6\gamma\theta-\theta^2}, \quad p_f^{WY*} = \\
& \frac{\gamma(4(1-\theta)-l(4-\gamma-3\theta))}{(8-\gamma)\gamma-6\gamma\theta-\theta^2}; \\
(ii) \quad & D_n^{WY*} = \frac{(1+l)\gamma-\theta+l\theta}{(8-\gamma)\gamma-6\gamma\theta-\theta^2}, D_f^{WY*} = \frac{\gamma(4-\gamma-2l(2-\theta)-3\theta)}{(8-\gamma)\gamma-6\gamma\theta-\theta^2}; \\
(iii) \quad & \pi_m^{WY*} = \frac{\gamma(l(l(2-\theta)+3\theta+\gamma-4)+2(1-\theta))}{(8-\gamma)\gamma-6\gamma\theta-\theta^2}, \pi_p^{WY*} = \frac{\gamma(1-\theta)((1+l)\gamma+(-1+l)\theta)^2}{((8-\gamma)\gamma-6\gamma\theta-\theta^2)^2}.
\end{aligned}$$

Lemma 1 delivers an important insight. Under the wholesale contract without pay-on-delivery, a longer waiting time (i.e., lower θ) unambiguously weakens the online channel. To sustain online demand, the platform must lower the online retail price, which compresses its own margin and, through the wholesale price, also reduces the manufacturer's profit (i.e., $\frac{\partial D_n^{WN*}}{\partial \theta} > 0$, $\frac{\partial w^{WN*}}{\partial \theta} > 0$, $\frac{\partial p_n^{WN*}}{\partial \theta} > 0$, $\frac{\partial \pi_m^{WN*}}{\partial \theta} > 0$). By contrast, once the pay-on-delivery service is introduced, delayed receipt no longer affects the online channel only through a lower product valuation. It also creates a payment-timing benefit for consumers, because payment is postponed until delivery. As a result, longer waiting time does not necessarily harm either the platform or the manufacturer under the WY scenario (i.e., $\frac{\partial \pi_m^{WY*}}{\partial \theta} < 0$ and $\frac{\partial \pi_p^{WY*}}{\partial \theta} < 0$ can be satisfied). The disadvantage of delayed receipt can be partly offset by the increased attractiveness of deferred payment.

Proposition 1 characterizes when the platform is willing to provide the pay-on-delivery service.

Proposition 1. *Under a wholesale contract, the platform will provide the pay-on-delivery service when $\theta < \gamma$.*

Proposition 1 demonstrates that the platform should optimize its strategy regarding the adoption of pay-on-delivery service by carefully evaluating the relative magnitudes of the value discount factor due to receipt delay (i.e., θ) and the revenue discount factor due to cash-flow opportunity cost (i.e., γ). With a small valuation discount factor, which indicates a prolonged online waiting time, the consumers' valuation of online products significantly decreases. Offering a pay-on-delivery service allows consumers to benefit from deferred payment, thereby mitigating the drawbacks of the online channel. However, this also incurs an opportunity cost for the platform. When $\theta < \gamma$, the benefit from the increased consumer utilities outweighs the platform's opportunity cost, enabling the platform to benefit from implementing the pay-on-delivery service. From Proposition 1, it is evident that in markets with long waiting times, adopting a pay-on-delivery service can be a profitable strategy to attract consumers. Additionally, effective cash flow management strategies should be in place to ensure the platform's financial stability.

We next compare the equilibrium outcomes under WN and WY to understand how the pay-on-delivery service reshapes channel competition.

Proposition 2. *After the platform offers the pay-on-delivery service under a wholesale contract ($\theta < \gamma$):*

- (i) The wholesale price, online retail price, and offline retail price increase ($w^{WY*} > w^{WN*}$, $p_n^{WY*} > p_n^{WN*}$, $p_f^{WY*} > p_f^{WN*}$).
- (ii) The online demand increases while the offline demand decreases ($D_n^{WY*} > D_n^{WN*}$, $D_f^{WY*} < D_f^{WN*}$).
- (iii) When $\theta \leq \frac{2(2-\gamma)-l(4+\gamma)}{2-l}$, the total demand increases ($D_n^{WY*} + D_f^{WY*} \geq D_n^{WN*} + D_f^{WN*}$); when $\theta > \frac{2(2-\gamma)-l(4+\gamma)}{2-l}$, the total demand decreases ($D_n^{WY*} + D_f^{WY*} < D_n^{WN*} + D_f^{WN*}$).

Proposition 2 reveals that, under the wholesale contract, pay-on-delivery raises both the online and offline retail prices, as well as the wholesale price. The intuition is straightforward. By allowing consumers to postpone payment, the service increases their effective willingness to pay in the online channel. The platform can therefore charge a higher online retail price. Anticipating this, the manufacturer also raises the wholesale price and the offline retail price. In this sense, pay-on-delivery softens price competition between the two channels. We refer to this mechanism as the *competition mitigation effect*.

At the same time, pay-on-delivery reallocates demand across channels. The online channel becomes more attractive, so online demand increases, whereas offline demand declines. We refer to this reallocation as the *channel cannibalization effect* of the pay-on-delivery service. Moreover, the increase in online demand does not necessarily dominate the loss in offline demand. Accordingly, total demand may either increase or decrease after pay-on-delivery is introduced.

We now turn to the impact of pay-on-delivery on profit and welfare under the wholesale contract.

Proposition 3. *After the platform offers the pay-on-delivery service under a wholesale contract ($\theta < \gamma$):*

- (i) The manufacturer always achieves a higher profit ($\pi_m^{WY*} > \pi_m^{WN*}$).
- (ii) Consumer surplus always decreases ($CS^{WY*} < CS^{WN*}$).
- (iii) Social welfare always increases ($SW^{WY*} > SW^{WN*}$).

Proposition 3 shows that the manufacturer always benefits when the platform adopts the pay-on-delivery service under the wholesale contract, even though the manufacturer does not directly transact with online consumers in this scenario. This result follows from the interaction of the two mechanisms described above. The *channel cannibalization effect* shifts demand away from the offline channel and is therefore unfavorable to the manufacturer. However, as shown in Proposition 2, the *competition mitigation effect* allows the manufacturer to raise both the wholesale price and the offline retail price. Because the manufacturer does not bear the cash-flow opportunity cost of delayed consumer payment under the wholesale contract, the gain from softer channel competition dominates the loss from cannibalization. As a result, the manufacturer is always better off under *WY* than under *WN*.

Proposition 3 also indicates that the pay-on-delivery service always reduces consumer surplus but increases social welfare under the wholesale contract. Although consumers benefit from paying later, this gain is more than offset by the higher retail prices induced by weaker channel competition. In other words, the payment-timing benefit is largely appropriated by firms through higher prices. At the same time, total firm profit rises sufficiently to outweigh the reduction in consumer surplus, leading to an increase in social welfare within the model.

4.2. Adoption of pay-on-delivery service under the agency contract

In this subsection, we examine the platform's decision to offer the pay-on-delivery service and the manufacturer's decision to adopt it under the agency contract. Compared with the wholesale case, the agency setting introduces an additional layer of strategic complexity because the implementation of pay-

on-delivery requires the consent of both parties. We first determine the equilibrium outcomes in AN and AY scenarios, then derive the conditions under which the service is offered and adopted, and finally compare the two scenarios to understand the operational consequences of pay-on-delivery service. The following lemma summarizes equilibrium solutions in both AN and AY scenarios.

Lemma 2. *In AN scenario, the equilibrium solutions are as follows:*

$$\begin{aligned} (i) \quad p_n^{AN*} &= \frac{\theta((2-\beta)(1-\theta)-l\beta)}{4(1-\beta)-(2-\beta)^2\theta}, \quad p_f^{AN*} = \frac{(1-\beta)(2(1-\theta)-l(2-\beta)\theta)}{4(1-\beta)-(2-\beta)^2\theta}, \\ (ii) \quad D_n^{AN*} &= \frac{l(2-\beta)-\beta}{4(1-\beta)-(2-\beta)^2\theta}, \quad D_f^{AN*} = \frac{(1-\beta)(2(1-l)-(2-\beta)\theta)}{4(1-\beta)-(2-\beta)^2\theta}, \\ (iii) \quad \pi_m^{AN*} &= \frac{(1-\beta)((1-l)^2-(1-l(2-\beta))\theta)}{4(1-\beta)-(2-\beta)^2\theta}, \quad \pi_p^{AN*} = \frac{\beta\theta(\beta-l(2-\beta))(l\beta-(2-\beta)(1-\theta))}{((2-\beta)^2\theta-4(1-\beta))^2}. \end{aligned}$$

In AY scenario, the equilibrium solutions are as follows:

$$\begin{aligned} (i) \quad p_n^{AY*} &= \frac{(1-\alpha-\beta)\gamma(1+l-\theta)+\theta(1-l-\theta)}{4(1-\alpha-\beta)\gamma-((1-\alpha-\beta)\gamma+\theta)^2}, \quad p_f^{AY*} = \frac{(1-\alpha-\beta)\gamma(2(1-\theta)-l(2-(1-\alpha-\beta)\gamma-\theta))}{4(1-\alpha-\beta)\gamma-((1-\alpha-\beta)\gamma+\theta)^2}, \\ (ii) \quad D_n^{AY*} &= \frac{(1+l)(1-\alpha-\beta)\gamma-\theta(1-l)}{4(1-\alpha-\beta)\gamma-((1-\alpha-\beta)\gamma+\theta)^2}, \quad D_f^{AY*} = \frac{(1-\alpha-\beta)\gamma(2(1-l)-(1-\alpha-\beta)\gamma-\theta)}{4(1-\alpha-\beta)\gamma-((1-\alpha-\beta)\gamma+\theta)^2}, \\ (iii) \quad \pi_m^{AY*} &= \frac{(1-\alpha-\beta)\gamma(1-\theta-l(2-l-(1-\alpha-\beta)\gamma-\theta))}{4(1-\alpha-\beta)\gamma-((1-\alpha-\beta)\gamma+\theta)^2}, \\ \pi_p^{AY*} &= \frac{(\alpha+\beta)\gamma((1+l)(1-\alpha-\beta)\gamma-\theta(1-l))((1-\alpha-\beta)\gamma(1+l-\theta)+\theta(1-l-\theta))}{(4(1-\alpha-\beta)\gamma-((1-\alpha-\beta)\gamma+\theta)^2)^2}. \end{aligned}$$

Lemma 2 highlights an important contrast with the wholesale setting. Without pay-on-delivery, a longer waiting time (i.e., lower θ) again weakens the online channel and hurts both parties (i.e., $\frac{\partial \pi_m^{AN*}}{\partial \theta} > 0$, $\frac{\partial \pi_p^{AN*}}{\partial \theta} > 0$). However, once pay-on-delivery is adopted, the consequences are no longer symmetric. In scenario AY , the manufacturer can benefit from a longer waiting time even though delayed receipt lowers the valuation of the online product (i.e., $\frac{\partial \pi_m^{AY*}}{\partial \theta} < 0$). The reason is that, under the agency contract, the manufacturer controls prices in both channels. A longer waiting time increases product differentiation between the online and offline channels, and the availability of deferred payment partly restores the attractiveness of the online channel. The manufacturer can therefore use its pricing authority to better segment the market and extract value from both channels.

We now derive the conditions under which pay-on-delivery is offered and adopted under the agency contract.

Proposition 4. *Under the agency contract, the platform will offer the pay-on-delivery service when $\theta < \theta_1$. Additionally, the manufacturer will adopt the service when $\theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$, where $\theta_1 > \frac{1-\alpha-\beta}{1-\beta}\gamma$.*

Proposition 4 shows that, different from the wholesale contract, the introduction of pay-on-delivery under the agency contract is constrained not only by the platform's willingness to provide the service but also by the manufacturer's willingness to adopt it. The platform benefits from pay-on-delivery over a relatively broad range because higher online demand increases commission-based revenue. The manufacturer, however, faces stricter conditions. Under the agency contract, the manufacturer already controls pricing in both channels and can coordinate online and offline prices to its advantage even without pay-on-delivery. Once the service is adopted, the manufacturer must bear both the cash-flow opportunity cost of delayed payment and the service fee charged by the platform. Therefore, the manufacturer adopts

pay-on-delivery only when the service sufficiently increases consumers' willingness to buy online, that is, when θ is sufficiently low.

This result yields a clear managerial implication. Even if the platform is willing to make pay-on-delivery available, a manufacturer operating under the agency contract may rationally decline to adopt it. The service is most attractive when delayed receipt poses a substantial disadvantage for the online channel, because only in that case is the demand-side benefit strong enough to compensate for the additional financial burden borne by the manufacturer.

We next compare the equilibrium outcomes under AN and AY .

Proposition 5. *Under the agency contract, when the platform offers the pay-on-delivery service and the manufacturer adopts it ($\theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$):*

- (i) *The online retail price always increases ($p_n^{AY*} > p_n^{AN*}$); when $\theta \leq \theta_2$ the offline retail price increases ($p_f^{AY*} \geq p_f^{AN*}$), and when $\theta > \theta_2$ the offline retail price decreases ($p_f^{AY*} < p_f^{AN*}$).*
- (ii) *The online demand increases while the offline demand decreases ($D_n^{AY*} > D_n^{AN*}$, $D_f^{AY*} < D_f^{AN*}$).*
- (iii) *Total demand increases ($D_f^{AY*} + D_n^{AY*} > D_f^{AN*} + D_n^{AN*}$).*

Proposition 5 shows that the impact of pay-on-delivery under the agency contract differs markedly from that under the wholesale contract. The online retail price always rises after the service is adopted, reflecting the higher willingness to pay induced by deferred payment. By contrast, the offline retail price does not always move in the same direction. It increases only when θ is sufficiently low and declines otherwise. This difference reflects the fact that, under the agency contract, the manufacturer internalizes both channels and can strategically reallocate profit between them. When θ is sufficiently low, the two channels are already well differentiated because delayed receipt makes the online channel relatively less attractive. In this case, pay-on-delivery further improves the online channel without intensifying cannibalization too much, allowing the manufacturer to raise prices in both channels. However, when θ is not sufficiently low, the channels are more directly substitutable. Because the manufacturer earns directly from both channels under agency selling, it may prefer to lower the offline retail price and shift demand toward the online channel, rather than soften competition across both channels.

Under the agency contract, pay-on-delivery still has a clear channel cannibalization effect. Online demand always rises and offline demand always falls, indicating persistent channel cannibalization. At the same time, adopting the pay-on-delivery service consistently raises the total demand. We refer to this as the *market expansion effect* under the agency contract. Relative to the wholesale case, pay-on-delivery is therefore more likely to enlarge the total market when pricing is coordinated by the manufacturer.

We now examine the implications for consumer surplus and social welfare.

Proposition 6. *Under the agency contract, when the platform offers the pay-on-delivery service and the manufacturer adopts it ($\theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$):*

- (i) *When $\theta \leq \theta_3$, consumer surplus decreases ($CS^{AY*} \leq CS^{AN*}$); when $\theta_3 < \theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$, consumer surplus increases ($CS^{AY*} > CS^{AN*}$).*
- (ii) *The social welfare always increases ($SW^{AY*} > SW^{AN*}$).*

Unlike the scenario depicted in Proposition 3 under a wholesale contract, consumers can benefit from the pay-on-delivery service under the agency contract. This benefit arises from the reduced competition mitigation effect under the agency contract. When θ is not sufficiently low, the manufacturer may reduce

the offline retail price while expanding the online channel, which can improve consumer surplus. By contrast, when θ is sufficiently low, the manufacturer raises prices more aggressively across channels, and the gain from deferred payment is no longer enough to offset the resulting price effect. Consumer surplus then declines.

Taken together, Proposition 6 shows that the welfare implications of pay-on-delivery under the agency contract are more nuanced than under the wholesale contract. The service still improves overall efficiency within the model, but its benefit to consumers depends on how strongly it relaxes online-channel disadvantage relative to how much pricing power the manufacturer can exercise after adoption.

4.3. Manufacturer's selection of distribution contract

In this subsection, we examine the manufacturer's equilibrium choice of distribution contract. Building on the adoption conditions derived in Sections 4.1 and 4.2, we compare the manufacturer's profits across the four scenarios, namely WN , WY , AN , and AY , to determine which combination of contract form and pay-on-delivery decision emerges in equilibrium. The results are summarized in Proposition 7.

Proposition 7. *The equilibrium distribution contract and pay-on-delivery service adoption are as follows:*

- (i) *If $(\theta \leq \frac{1-\alpha-\beta}{1-\beta}\gamma$ and $\beta > \beta_1$) or $(\frac{1-\alpha-\beta}{1-\beta}\gamma < \theta < \gamma$ and $\beta > \beta_2)$, the equilibrium scenario is WY .*
- (ii) *If $\theta \geq \gamma$ and $\beta > \beta_3$, the equilibrium scenario is WN .*
- (iii) *If $0 < \theta \leq \frac{1-\alpha-\beta}{1-\beta}\gamma$ and $\beta \leq \beta_1$, the equilibrium scenario is AY .*

Otherwise, the equilibrium scenario is AN .

For ease of exposition, Fig.3. illustrates the equilibrium regions implied by Proposition 7.

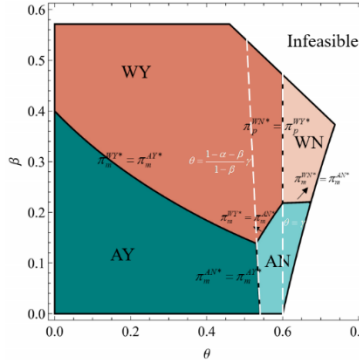


Fig.3. Equilibrium scenarios ($\gamma = 0.6$, $l = 0.4$, $\alpha = 0.1$).

Proposition 7 shows that the manufacturer's equilibrium choice is jointly determined by the commission rate of the agency contract β and the valuation discount factor due to receipt delay θ . Intuitively, the manufacturer prefers the agency contract when β is low and opts for the wholesale contract when β is high. A lower commission rate enables the manufacturer to retain a larger share of online revenue under agency contract, whereas a higher commission rate makes wholesale contract more attractive by reducing the burden of platform revenue sharing. The role of θ , however, is more nuanced. The equilibrium regions in Proposition 7 can be understood by dividing the parameter space into three cases.

First, when θ is sufficiently low, namely $\theta \leq \frac{1-\alpha-\beta}{1-\beta}\gamma$, the pay-on-delivery service is adopted under both distribution contracts. In this region, waiting time is sufficiently long, so deferred payment creates substantial value for consumers under both contracts. The equilibrium choice is therefore between AY and

WY. When β is low (i.e., $\beta \leq \beta_1$), the manufacturer prefers AY, because it can directly control retail prices in both channels and capture more of the increased online willingness to pay. When β is high (i.e., $\beta > \beta_1$), the manufacturer prefers WY, because the agency commission erodes the gain from online channel and makes the wholesale contract more attractive. Thus, in the low- θ region, Fig. 3 is best interpreted as showing how the commission rate determines whether the manufacturer prefers to internalize the online channel through agency contract or delegate it to the platform through wholesale contract.

Second, when $\frac{1-\alpha-\beta}{1-\beta}\gamma < \theta < \gamma$, the pay-on-delivery service will be adopted only under the wholesale contract. This is the most interesting region, because the payment scheme itself begins to reshape the contract choice. According to Proposition 4, under the agency contract, the manufacturer no longer finds it profitable to adopt pay-on-delivery. Conversely, according to Proposition 3, the manufacturer is always better off under WY than under WN. Therefore, the relevant comparison becomes WY and AN. When β is high (i.e., $\beta > \beta_2$), the manufacturer prefers WY, because it can still benefit indirectly from the platform's provision of pay-on-delivery without directly bearing the associated financial burden. Otherwise, when β is low, the manufacturer prefers AN, because agency contract allows it to exploit direct pricing authority once the online waiting-time disadvantage becomes less severe. As θ rises within this interval, the waiting-time disadvantage of the online channel weakens, the value of deferred payment declines, and the relative advantage of AN becomes stronger. Consequently, the WY region shrinks and the AN region expands.

Third, when $\theta \geq \gamma$, pay-on-delivery is not provided under either contract, the comparison reduces to WN versus AN. In this region, the online channel is no longer sufficiently disadvantaged for payment deferral to matter. The contract choice is therefore driven primarily by the standard trade-off between direct selling under agency and double marginalization under wholesale. When β is low, the manufacturer prefers AN, because it can directly capture online margin and coordinate pricing across the two channels. When β is high, the manufacturer prefers WN, because agency selling becomes too expensive.

To further isolate the role of pay-on-delivery in the contract choice, we next compare the equilibrium contract under the benchmark without pay-on-delivery and the equilibrium contract when pay-on-delivery is feasible. Proposition 8 summarizes how the service changes the manufacturer's distribution-contract preference.

Proposition 8. *The pay-on-delivery service has a substantial influence on the manufacturer's selection of distribution contracts. Compared to the scenario without pay-on-delivery service, when the pay-on-delivery service is considered:*

- (i) *If $\alpha < \alpha_0$ and $0 < \theta \leq \theta_4$, the manufacturer is more likely to choose the agency contract;*
- (ii) *If $\{\alpha < \alpha_0 \text{ and } \theta_4 < \theta < \gamma\}$ or $\{\alpha \geq \alpha_0 \text{ and } \theta < \gamma\}$ the manufacturer is more likely to choose the wholesale contract.*

For ease of description, we have illustrated Fig.4. based on Proposition 8.

Proposition 8 clarifies that pay-on-delivery affects contract choice through two countervailing forces. On the one hand, it raises the attractiveness of the online channel by allowing consumers to postpone payment, which tends to favor the agency contract because the manufacturer can directly exploit its pricing authority. On the other hand, under the agency contract the manufacturer must bear delayed revenue realization and also pay the service fee α , which tends to favor the wholesale contract.

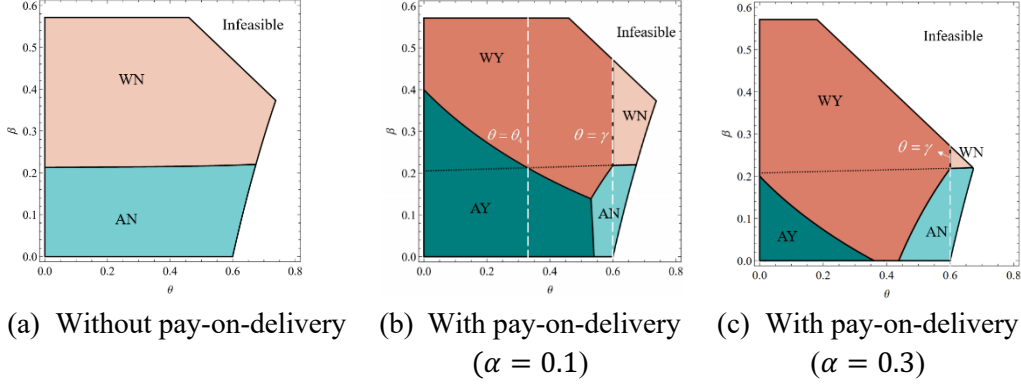


Fig.4. Equilibrium Scenarios ($\gamma = 0.6, l = 0.4$).

When α is sufficiently high (i.e., $\alpha \geq \alpha_0$), the second force dominates. In this case, the manufacturer becomes more likely to choose the wholesale contract once pay-on-delivery is considered. The intuition is straightforward. Under the agency contract, adopting pay-on-delivery is financially costly because the manufacturer bears both the timing-related burden and the additional service fee. Under the wholesale contract, by contrast, the manufacturer can still benefit indirectly from the platform's adoption of pay-on-delivery without directly paying for the service. Therefore, a sufficiently high α systematically shifts the manufacturer toward the wholesale contract.

When α is relatively low (i.e., $\alpha < \alpha_0$), the impact of pay-on-delivery depends on θ . If $0 < \theta \leq \theta_4$, the manufacturer becomes more likely to choose the agency contract. In this region, waiting time is long and the online channel is heavily disadvantaged without payment deferral. Pay-on-delivery therefore creates substantial value for consumers, and the manufacturer can capture this value more effectively under the agency contract by setting both online and offline retail prices. The gain from stronger pricing flexibility outweighs the direct financial burden of adoption, making the agency contract relatively more attractive.

By contrast, when $\theta_4 < \theta < \gamma$, the manufacturer becomes more likely to choose the wholesale contract once pay-on-delivery is taken into account. Here, the waiting-time disadvantage remains relevant but is no longer severe enough to justify bearing the direct cost of adoption under the agency contract. The manufacturer therefore has a stronger incentive to let the platform provide the service and to benefit from it indirectly under wholesale selling. In this sense, pay-on-delivery does not uniformly favor one contract over the other. Its effect on governance depends on the balance between the demand-side value of deferred payment and the financial burden of direct adoption.

Overall, Propositions 7 and 8 jointly show that pay-on-delivery is not merely a payment arrangement. It changes the relative value of wholesale and agency selling by altering channel attractiveness, reallocating delayed-cash-flow exposure, and changing how much value the manufacturer can extract through pricing authority.

5. Extensions

In this section, we present two extensions to assess the robustness of the main model. First, we allow for product returns in the online channel and examine how the return-related benefit of pay-on-delivery affects the equilibrium outcomes. Second, we relax the assumption that consumers apply the same discount factor to product valuation and to payment, and instead allow the two factors to differ. Together, these extensions show that the central insights of the paper continue to hold once additional operational features of pay-on-delivery are taken into account.

5.1. Considering product returns

In this subsection, we consider the product return behavior of consumers. In the offline channel, consumers can inspect the product before purchase and therefore choose the product that best fits their needs. The offline utility remains $u_f = v - p_f - l$. In the online channel, by contrast, consumers cannot inspect the product prior to purchase. Let $\delta \in (0,1)$ denote the probability that the product received by the consumer matches the consumer's needs. Accordingly, $1 - \delta$ is the probability of a mismatch. When a mismatch occurs, the consumer may return the product after delivery. The pay-on-delivery service significantly reduces the hassle associated with returns because consumers can inspect the product at delivery and immediately refuse it if it is unsuitable. We therefore normalize the return hassle cost under pay-on-delivery to zero. Without pay-on-delivery, returning an unsuitable product entails a positive hassle cost $m > 0$. Consequently, the utility of purchasing online with the pay-on-delivery service is $u_n^{Y1} = \delta\theta(v - p_n)$, where the superscript 1 denotes the return-related case. The utility of purchasing online without pay-on-delivery service is $u_n^{N1} = \delta(\theta v - p_n) - (1 - \delta)m$. We then derive the demand functions based on these utilities.

$$D_f = \begin{cases} 1 - \frac{p_f - \delta p_n + l - m(1 - \delta)}{1 - \delta\theta}, & \text{without the pay-on-delivery service} \\ 1 - \frac{p_f - \delta\theta p_n + l}{1 - \delta\theta}, & \text{with the pay-on-delivery service} \end{cases}$$

$$D_n = \begin{cases} \frac{p_f - \delta p_n + l - m(1 - \delta)}{1 - \delta\theta} - \frac{\delta p_n + m(1 - \delta)}{\delta\theta}, & \text{without the pay-on-delivery service} \\ \frac{p_f - \delta\theta p_n + l}{1 - \delta\theta} - p_n, & \text{with the pay-on-delivery service} \end{cases}$$

Once a product is returned, the party that sold the product to the consumer incurs a unit return-handling cost c , which captures costs such as repackaging and resale processing (Y. Li et al., 2023). Under the wholesale contract, this cost is borne by the platform; under the agency contract, it is borne by the manufacturer.

We define Scenarios *WNI/WYI* as cases in which the manufacturer opts for a wholesale contract without/with pay-on-delivery, and Scenarios *ANI/AYI* as cases in which the manufacturer opts for an agency contract without/with pay-on-delivery. The profit functions of the firms under four scenarios are shown in Table 3.

Table 3 Profit functions considering product returns.

Scenarios	Profit functions
<i>WNI</i>	$\pi_m^{WN1} = p_f^{WN1} D_f^{WN1} + w^{WN1} D_n^{WN1},$
	$\pi_p^{WN1} = \delta p_n^{WN1} D_n^{WN1} - (1 - \delta)c D_n^{WN1} - w^{WN1} D_n^{WN1}.$
<i>WYI</i>	$\pi_m^{WY1} = p_f^{WY1} D_f^{WY1} + w^{WY1} D_n^{WY1},$
	$\pi_p^{WY1} = \delta \gamma p_n^{WY1} D_n^{WY1} - (1 - \delta)c D_n^{WY1} - w^{WY1} D_n^{WY1}$
<i>ANI</i>	$\pi_m^{AN1} = p_f^{AN1} D_f^{AN1} + \delta(1 - \beta)p_n^{AN1} D_n^{AN1} - (1 - \delta)c D_n^{AN1},$
	$\pi_p^{AN1} = \delta\beta p_n^{AN1} D_n^{AN1}$
<i>AYI</i>	$\pi_m^{AY1} = p_f^{AY1} D_f^{AY1} + \delta(1 - \alpha - \beta)\gamma p_n^{AY1} D_n^{AY1} - (1 - \delta)c D_n^{AY1},$
	$\pi_p^{AY1} = \delta(\alpha + \beta)\gamma p_n^{AY1} D_n^{AY1}$

Using backward induction, the equilibrium solutions in four scenarios are derived. We compare the manufacturer's profits across the four scenarios to determine which combination of contract form and pay-on-delivery decision emerges in equilibrium. The results are summarized in Proposition 9.

Proposition 9. *When product return is considered, the equilibrium distribution contract and pay-on-delivery service adoption are as follows:*

- (i) *If $(\theta \leq \theta_5 \text{ and } \beta > \beta_4)$ or $(\theta_5 < \theta < \theta_6 \text{ and } \beta > \beta_5)$, the equilibrium scenario is WY1.*
- (ii) *If $\theta \geq \theta_6$ and $\beta > \beta_6$, the equilibrium scenario is WN1.*
- (iii) *If $0 < \theta \leq \theta_5$ and $\beta \leq \beta_4$, the equilibrium scenario is AY1.*
- (iv) *Otherwise, the equilibrium scenario is AN1.*

For ease of description, we have illustrated Fig.5 based on Proposition 9.

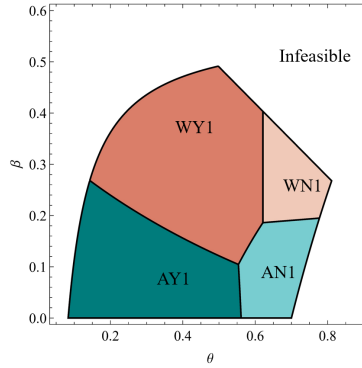


Fig.5 Equilibrium scenarios with product returns.

($\gamma = 0.6, l = 0.4, \alpha = 0.1, c = 0.2, \delta = 0.9, m = 0.1$)

Proposition 9 demonstrates that, when accounting for product return behavior, the pay-on-delivery service will be implemented under both contracts when the valuation discount factor due to delayed delivery is relatively small (i.e., waiting time is long). This finding aligns with the insights derived from Proposition 1 and Proposition 4. More importantly, the equilibrium regions remain organized around the same logic as in the main model: the attractiveness of pay-on-delivery is governed by the trade-off between its demand-enhancing benefit and the financial burden it creates, while the manufacturer's contract choice continues to depend on the balance between pricing authority and cost exposure. The extension therefore confirms that the main equilibrium logic is robust to return-related considerations. In Proposition 10, the effect of product return behavior on the adoption of the pay-on-delivery service and the selection of distribution contracts are examined.

Proposition 10. *When product returns are considered, as opposed to scenarios without product returns:*

- (i) *The pay-on-delivery service is more likely to be adopted, even though it leads to a higher return volume.*
- (ii) *The manufacturer is more likely to choose the wholesale contract.*

Proposition 10 delivers two additional insights. First, once product returns are considered, pay-on-delivery becomes more likely to be adopted, even though it leads to a higher return volume. The intuition is straightforward. Without pay-on-delivery, consumers anticipate the hassle cost of returning mismatched products and therefore become more reluctant to purchase online. By contrast, pay-on-delivery eliminates this return hassle at the point of delivery, which increases consumers' willingness to purchase. This demand-side benefit can dominate the additional return-related cost, making pay-on-delivery more attractive than in the main model.

Second, the manufacturer becomes more likely to choose the wholesale contract when returns are taken into account. Under wholesale selling, the manufacturer does not bear the return-handling cost and can still benefit indirectly from the stronger online channel through higher wholesale and offline prices. By contrast, under agency selling, the manufacturer directly bears return-related losses once consumers purchase online and later return the product. As a result, introducing return behavior strengthens the relative attractiveness of the wholesale contract.

5.2. Different discount factors for product valuation and cash

In the main model, we assume that the consumers' utility of buying online with pay-on-delivery service is $u_n^Y = \theta(v - p_n)$, where we assume that the consumers' discount factor for product valuation and cash are equivalent. This assumption captures the case in which payment occurs exactly when the product is received. In practice, however, consumers may sometimes have different discount factors for products and cash-flow even when the product is paid upon receipt. For example, fresh products such as flowers and fruits can quickly perish, so the discount factor for fresh products is lower than that for cash. For non-essential items, consumers might not mind waiting for receipts because they do not require immediate use, but they might value their cash for other potential needs. Therefore, the discount factor of cash is lower than that for non-essential products.

To capture this possibility, we use different parameters to capture the discount factor for products and cash. When buying the product online with pay-on-delivery service, the utility of consumers will be $u_n^{Y2} = \theta v - \theta' p_n$, with $\theta' = \eta\theta$, $\eta > 0$ and $\eta \neq 1$. If $0 < \eta < 1$, the discount factor of cash is lower than that of the product; if $\eta > 1$, it is higher. The superscript 2 denotes the case in this extension. The utility function for consumers purchasing offline remains $u_f = v - p_f - l$, and the profit functions are likewise unchanged. We define Scenarios *WN2/WY2* as cases in which the manufacturer opts for a wholesale contract without/with pay-on-delivery, and Scenarios *AN2/AY2* as cases in which the manufacturer opts for an agency contract without/with pay-on-delivery.

The conditions under which the platform will opt to offer the pay-on-delivery service are presented in Proposition 11.

Proposition 11. *When product valuation and cash are subject to different discount factors:*

- (i) *Under the wholesale contract, the platform will provide the pay-on-delivery service when $\theta < \frac{\gamma}{\eta}$.*
- (ii) *Under the agency contract, the platform will offer the pay-on-delivery service and the manufacturer will adopt it when $\theta < \frac{1-\alpha-\beta}{(1-\beta)\eta}\gamma$.*

Proposition 11 demonstrates that the pay-on-delivery service will be offered when the discount factor for product valuation is sufficiently low, which is consistent with the fundamental insights derived from the main model. Additionally, Proposition 11 highlights that when η increases, the pay-on-delivery service is less likely to be offered. This is because a higher η amplifies the cash discount factor ($\eta\theta$), meaning that consumers perceive a diminished benefit from the delayed payment. The financial flexibility provided by the pay-on-delivery service is less likely to compensate for the utility lost due to the delayed product receipt. Consequently, the service loses its appeal to consumers, ultimately reducing the platform's or manufacturer's incentive to provide this service.

We further investigate the equilibrium conditions regarding the distribution contract and pay-on-delivery service adoption, as detailed in Proposition 12.

Proposition 12. *When product valuation and cash are subject to different discount factors, the equilibrium distribution contract and pay-on-delivery service adoption are as follows:*

- (i) *If $(\theta \leq \frac{1-\alpha-\beta}{(1-\beta)\eta}\gamma$ and $\beta > \beta_7$) or $(\frac{1-\alpha-\beta}{(1-\beta)\eta}\gamma < \theta < \frac{\gamma}{\eta}$ and $\beta > \beta_8$), the equilibrium scenario is WY2.*
- (ii) *If $\theta \geq \frac{\gamma}{\eta}$ and $\beta > \beta_9$, the equilibrium scenario is WN2.*
- (iii) *If $0 < \theta \leq \frac{1-\alpha-\beta}{(1-\beta)\eta}\gamma$ and $\beta \leq \beta_7$, the equilibrium scenario is AY2.*
- (iv) *Otherwise, the equilibrium scenario is AN2.*

For ease of description, we have illustrated the equilibrium scenarios in Fig. 6.

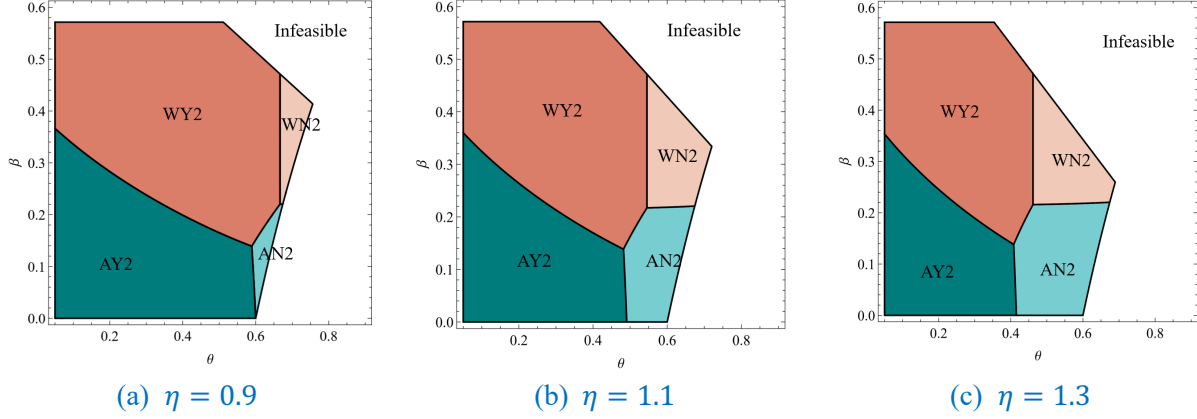


Fig.6. Equilibrium Scenarios ($\gamma = 0.6, l = 0.4, \alpha = 0.1$).

Proposition 12 shows that allowing different discount factors for product valuation and cash-flow changes the threshold values but does not overturn the equilibrium logic of the main model, strengthening the robustness of the main analysis. The same four scenarios remain possible, and the equilibrium regions continue to be governed by the interaction among delivery-delay disadvantage, payment-timing benefit, revenue discount, and contractual structure. Additionally, η acts as a simple and intuitive modifier of the strength of the pay-on-delivery mechanism. A lower η increases the relative value of deferred payment and therefore enlarges the parameter region in which pay-on-delivery is attractive. A higher η compresses that region and limits the benefit of pay-on-delivery service.

6. Conclusion

The pay-on-delivery service is widely embraced by e-commerce platforms as it allows consumers to pay upon receipt, mitigating the disadvantage of long waiting times under both wholesale and agency contracts. However, the interaction between pay-on-delivery services and distribution contracts remains unexplored. In this paper, we present a stylized model of a dual-channel supply chain, featuring an upstream manufacturer with an offline channel and an e-commerce platform. The central insight is that pay-on-delivery is not merely a payment convenience. By postponing consumer payment until delivery, it simultaneously improves the attractiveness of the online channel and delays firms' cash recovery. Its value therefore depends on the balance between the demand-side benefit of deferred payment and the financial burden of postponed revenue realization.

Our analysis first sheds light on the conditions under which pay-on-delivery is adopted. Under the wholesale contract, the platform offers the service when the benefit from alleviating the online channel's waiting-time disadvantage outweighs the platform's cash-flow opportunity cost. Under the agency contract, adoption is more restrictive because the manufacturer must bear both delayed revenue realization and the

pay-on-delivery service fee. As a result, even when the platform is willing to provide the service, the manufacturer may rationally decline to adopt it unless the waiting-time disadvantage in the online channel is sufficiently severe.

The paper also shows that pay-on-delivery has important implications for channel outcomes. Under the wholesale contract, it softens price competition between the online and offline channels, reallocates demand toward the online channel, and always increases the manufacturer's profit indirectly. Under the agency contract, its effects are more nuanced. Although pay-on-delivery still shifts demand toward the online channel and expands the total market, it does not always benefit the manufacturer, because the manufacturer directly bears the timing-related burden and the service fee. These differences also translate into distinct welfare outcomes. Pay-on-delivery always improves social welfare within the model, but its effect on consumer surplus depends on the contractual arrangement and the severity of delivery delay.

A further contribution of the paper is to show how pay-on-delivery reshapes the manufacturer's contract choice. When the pay-on-delivery service fee is sufficiently high, the manufacturer becomes more likely to prefer the wholesale contract, because wholesale selling allows it to benefit from the service indirectly without directly bearing the associated financial burden. When the service fee is relatively low, the direction of this effect depends on the valuation discount factor due to receipt delay. If the waiting-time disadvantage is sufficiently severe, pay-on-delivery makes the agency contract more attractive because the manufacturer can better exploit pricing authority across the two channels. If the waiting-time disadvantage is moderate, pay-on-delivery instead strengthens the relative attractiveness of the wholesale contract.

Finally, we examine two extensions. When product returns are incorporated, pay-on-delivery becomes more likely to be adopted because it eliminates the return hassle perceived by consumers at the point of delivery, although it also leads to a higher return volume. When the discount factors for product valuation and payment differ, the equilibrium thresholds change, but the main logic of the main model remains intact. These findings suggest that the core mechanism identified in the paper is robust to richer operational features of pay-on-delivery.

We conclude the paper by proposing potential directions for future research. First, we examine a supply chain comprising an upstream manufacturer and a platform. The impact of competition between the manufacturers or the platforms can be explored in future research. Second, this research considers only two kinds of contracts. It may be interesting to study how other cooperation formats interact with the pay-on-delivery service. Third, the channels in our work are separately operated. Analyzing the situation incorporating the omnichannel may be interesting.

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Appendix

Proof of Lemma 1. and Lemma 2.

The proof of the scenario WN is as follows:

It is easy to know $\frac{\partial^2 \pi_p^{WN}}{\partial (p_n^{WN})^2} = -\frac{2}{(1-\theta)\theta} < 0$.

By $\frac{\partial \pi_p^{WN}}{\partial p_n^{WN}} = 0$, we have $p_n^{WN}(w^{WN}, p_f^{WN}) = \frac{1}{2}(w^{WN} + l\theta + \theta p_f^{WN})$. Substituting $p_n^{WN}(w^{WN}, p_f^{WN})$ into $\pi_m^{WN}(w^{WN}, p_f^{WN})$. The Hessian matrix of $\pi_m^{WN}(w^{WN}, p_f^{WN})$ is negative.

By $\frac{\partial \pi_m^{WN}(w^{WN}, p_f^{WN})}{\partial w^{WN}} = 0$, $\frac{\partial \pi_m^{WN}(w^{WN}, p_f^{WN})}{\partial p_f^{WN}} = 0$, we obtain $w^{WN*} = \frac{\theta}{2}$, $p_f^{WN*} = \frac{1-l}{2}$, $p_n^{WN*} = \frac{(2+l)\theta}{4}$.

The proof of the scenarios WY , AN , and AY is similar to that of scenario WN .

To ensure the Hessian matrixes are negative, $(-8 + \gamma)\gamma + 6\gamma\theta + \theta^2 < 0$, $4 - 4\beta - (-2 + \beta)^2\theta > 0$, $4(-1 + \alpha + \beta)\gamma + (-(-1 + \alpha + \beta)\gamma + \theta)^2 < 0$ and $(-1 + \alpha + \beta) < 0$ should be satisfied. Additionally, we concentrate on the interesting situations when the market is partially covered and the demand of two channels is non-negative, which satisfies $\max\{\frac{\beta}{2-\beta}, 1 + \frac{2(-1+\alpha+\beta)\gamma}{\gamma-\alpha\gamma-\beta\gamma+\theta}, 1 - \frac{2\gamma}{\gamma+\theta}\} < l < \min\{2 - \frac{2}{2-\theta}, \frac{4-\gamma-3\theta}{2(2-\theta)}, 1 - \frac{1}{2}(2 - \beta)\theta, \frac{1}{2}(2 - (1 - \alpha - \beta)\gamma - \theta)\}$, $\gamma > \frac{\theta(l+\theta-1)}{(1-\alpha-\beta)(l-\theta+1)}$, and $-(\theta(-1 + l + \theta) + \gamma(-3 - 3l + 3\theta + 2l\theta)) > 0$

Proof of Proposition 1.

$\pi_p^{WY*} - \pi_p^{WN*} = \frac{(\theta-\gamma)F}{16(1-\theta)(-8\gamma+\gamma^2+6\gamma\theta+\theta^2)^2}$, $F = (l^2\gamma^3\theta - l^2\theta^4 + \gamma\theta(16(-1+\theta)^2 - 32l(-1+\theta)^2 + l^2(-4+\theta)(-4+3\theta)) - \gamma^2(16(-1+\theta)^2 + 32l(-1+\theta)^2 + l^2(-4+\theta)(-4+3\theta)))$. F is a quadratic function about l .

When $\theta > \gamma$, $\frac{\partial^2 F}{\partial l^2} > 0$, $F(l=0) > 0$, and $\text{argmin} F(l) > 0$. Therefore, $F > 0$ when $l < l_1$ or $l > l_2$. l_1 and l_2 satisfies $F(l_1) = 0$, $F(l_2) = 0$, and $l_1 < l_2$. It can be proved that $l_1 < l_{\min}$, $l_2 > l_{\max}$. Consequently, $F < 0$ when $\theta > \gamma$.

When $\theta < \gamma$, $\frac{\partial^2 (-F)}{\partial l^2} > 0$, $-F(l=0) > 0$, and $\text{argmin} -F(l) < 0$. Therefore, $-F > 0$ always satisfied when $\theta < \gamma$.

To sum up, $\pi_p^{WY*} - \pi_p^{WN*} > 0$ when $\theta < \gamma$.

Proof of Proposition 2.

On condition that $\theta < \gamma$,

$$\frac{\partial (w^{WY*} - w^{WN*})}{\partial l} > 0; w^{WY*} - w^{WN*} > w^{WY*} - w^{WN*} \left(l = 1 - \frac{2\gamma}{\gamma+\theta} \right) > 0;$$

$$\frac{\partial (p_n^{WY*} - p_n^{WN*})}{\partial l} > 0; p_n^{WY*} - p_n^{WN*} > p_n^{WY*} - p_n^{WN*} (l=0) > 0;$$

$$\frac{\partial (p_f^{WY*} - p_f^{WN*})}{\partial l} > 0; p_f^{WY*} - p_f^{WN*} > p_f^{WY*} - p_f^{WN*} \left(l = 1 - \frac{2\gamma}{\gamma+\theta} \right) = 0;$$

$$\frac{\partial (D_n^{WY*} - D_n^{WN*})}{\partial l} < 0; D_n^{WY*} - D_n^{WN*} > D_n^{WY*} - D_n^{WN*} \left(l = \frac{4-\gamma-3\theta}{2(2-\theta)} \right) > 0;$$

$$\frac{\partial(D_f^{WY*} - D_f^{WN*})}{\partial l} < 0; D_f^{WY*} - D_f^{WN*} < D_f^{WY*} - D_f^{WN*}(l=0) < 0;$$

$$\frac{\partial((D_n^{WY*} + D_f^{WY*}) - (D_n^{WN*} + D_f^{WN*}))}{\partial l} < 0; \text{ when } l = \frac{-4+\gamma+3\theta}{2(-2+\theta)}, (D_n^{WY*} + D_f^{WY*}) - (D_n^{WN*} + D_f^{WN*}) = 0; \text{ so}$$

$$(D_n^{WY*} + D_f^{WY*}) - (D_n^{WN*} + D_f^{WN*}) > 0 \text{ when } l \leq \frac{4-2\gamma-2\theta}{4+\gamma-\theta} \text{ (i.e., } \theta \leq \frac{2(2-\gamma)-l(4+\gamma)}{2-l} \text{), and } (D_n^{WY*} +$$

$$D_f^{WY*}) - (D_n^{WN*} + D_f^{WN*}) < 0 \text{ when } l > \frac{4-2\gamma-2\theta}{4+\gamma-\theta} \text{ (i.e., } \theta > \frac{2(2-\gamma)-l(4+\gamma)}{2-l} \text{).}$$

Proof of Proposition 3.

Denote $G_1(l) = \pi_m^{WY*} - \pi_m^{WN*}$, which is a quadratic function about l . When $\theta < \gamma$, $\frac{\partial^2 G_1}{\partial l^2} > 0$, $G_1(l=0) > 0$, and $\text{argmin} G_1(l=0) > 0$. Therefore, $G_1 > 0$ is always satisfied when $\theta < \gamma$.

Denote $G_2(l) = CS^{WY*} - CS^{WN*}$, where $CS^{WN*} = \int_{p_r^*}^{\frac{p_d^*+l-p_r^*}{1-\theta}} u_n^{WN*} dv + \int_{\frac{p_d^*+l-p_r^*}{1-\theta}}^1 u_f^{WN*} dv$,

$CS^{WY*} = \int_{p_r^*}^{\frac{p_d^*+l-\theta p_r^*}{1-\theta}} u_n^{WN*} dv + \int_{\frac{p_d^*+l-\theta p_r^*}{1-\theta}}^1 u_f^{WN*} dv$. $G_2(l)$ is a quadratic function about l . When $\theta < \gamma$,

$\frac{\partial^2 G_2}{\partial l^2} > 0$, $G_2(l=0) < 0$. Therefore, $G_2 < 0$ when $l < l_m$ and $G_2 \geq 0$ when $l \geq l_m$, where l_m is the second root of $G_2(l) = 0$. It can be proved that $l_{max} < l_m$, so $G_2 < 0$ always satisfied.

Denote $G_3(l) = SW^{WY*} - SW^{WN*}$, where $SW^{k*} = \pi_m^{k*} + \pi_p^{k*} + CS^{k*}$. $G_3(l)$ is a quadratic function about l . When $\theta < \gamma$, $\frac{\partial^2 G_3}{\partial l^2} > 0$, $G_3(l=0) > 0$, and $\text{argmin} G_3(l=0) > 0$. Therefore, $G_3 > 0$ is always satisfied when $\theta < \gamma$.

Proof of Proposition 4.

Denote $F_A(l) = \pi_m^{AY*} - \pi_m^{AN*}$, which is a quadratic function about l .

When $\theta < (1-\alpha-\beta)(1-\beta)\gamma$, $\frac{\partial^2 F_A}{\partial l^2} > 0$, $F_A(l=0) > 0$, and $\text{argmin} F_A(l) > 0$. Therefore, $F_A > 0$ always satisfied when $\theta < (1-\alpha-\beta)(1-\beta)\gamma$;

When $(1-\alpha-\beta)(1-\beta)\gamma < \theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$, $\frac{\partial^2 F_A}{\partial l^2} < 0$, $F_A(l=0) > 0$, and $\text{argmax} F_A(l) > 0$. Therefore, $F_A > 0$ when $l_{1A} < l < l_{2A}$. l_{1A} and l_{2A} satisfies $F_A(l_{1A}) = 0$, $F_A(l_{2A}) = 0$, and $l_{1A} < l_{2A}$. It can be proved that $l_{1A} < l_{min}$ and $l_{2A} > l_{max}$. Consequently, $F_A > 0$ when $(1-\alpha-\beta)(1-\beta)\gamma < \theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$.

When $\theta > \frac{1-\alpha-\beta}{1-\beta}\gamma$, $\frac{\partial^2 F_A}{\partial l^2} > 0$, $F_A(l=0) > 0$, and $\text{argmin} F_A(l) < 0$. Therefore, $F_A > 0$ when $l < l_{1A}$ or $l > l_{2A}$. It can be proved that $l_{1A} < l_{min}$ and $l_{2A} > l_{max}$. Consequently, $F_A < 0$ when $\theta > \frac{1-\alpha-\beta}{1-\beta}\gamma$.

To sum up, $\pi_m^{AY*} - \pi_m^{AN*} > 0$ when $\theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$.

Similarly, we can find that when $\theta < \theta_1$, $\pi_p^{AY*} - \pi_p^{AN*} > 0$, where θ_1 is the first root of $\pi_p^{AY*} - \pi_p^{AN*} = 0$. $\theta_1 > \gamma > \frac{1-\alpha-\beta}{1-\beta}\gamma$.

Proof of Proposition 5.

It is similar to Proposition 2 so we omit it. Note that θ_2 is the first root of $p_f^{AY*} - p_f^{AN*} = 0$.

Proof of Proposition 6.

It is similar to Proposition 3 so we omit it. Note that θ_3 is the first root of $CS^{AY*} - CS^{AN*} = 0$.

Proof of Proposition 7.

(i) WY: $\pi_m^{WY*} > \pi_m^{AY*}$ when $\theta < \frac{1-\alpha-\beta}{1-\beta}\gamma$, or $\pi_m^{WY*} > \pi_m^{AN*}$ when $\frac{1-\alpha-\beta}{1-\beta}\gamma < \theta < \gamma$; $\pi_m^{WY*} >$

π_m^{AY*} equals to $\beta > \beta_1$, $\pi_m^{WY*} > \pi_m^{AN*}$ equals to $\beta > \beta_2$, where β_1 is the first root of $\pi_m^{WY*} = \pi_m^{AY*}$ and β_2 is the first root of $\pi_m^{WY*} = \pi_m^{AN*}$.

(ii) $WN: \pi_m^{WN*} > \pi_m^{AN*}$ and $\theta > \gamma$. $\pi_m^{WN*} > \pi_m^{AN*}$ equals to $\beta > \beta_3$, where β_3 is the first root of $\pi_m^{WN*} = \pi_m^{AN*}$.

(iii) $AY: \pi_m^{AY*} > \pi_m^{WY*}$ and $\theta < \frac{1-\alpha-\beta}{1-\beta} \gamma$. $\pi_m^{AY*} > \pi_m^{WY*}$ equals to $\beta < \beta_1$.

(iv) In other feasible regions, the equilibrium strategy is AN .

Proof of Proposition 8.

To prove Proposition 8, we need to prove that $\beta_2 \leq \beta_3$ when $\frac{1-\alpha-\beta}{1-\beta} \gamma \leq \theta \leq \gamma$ and if $\alpha \geq \alpha_0$, $\beta_1 \leq \beta_3$

when $0 \leq \theta \leq \frac{1-\alpha-\beta}{1-\beta} \gamma$; if $\alpha < \alpha_0$, (a) $\beta_1 > \beta_3$ when $\theta < \theta_2$ (b) $\beta_1 < \beta_3$ when $\theta_2 < \theta \leq \frac{1-\alpha-\beta}{1-\beta} \gamma$;

Note that $\frac{\partial \beta_1}{\partial \theta} < 0$, $\frac{\partial \beta_2}{\partial \theta} > \frac{\partial \beta_3}{\partial \theta} > 0$; when $\theta = \gamma$, $\beta_2 = \beta_3$, so when $\theta \leq \gamma$, $\beta_2 \leq \beta_3$.

If $\alpha \geq \alpha_0 = \frac{3}{4} - \frac{2l^2}{l(2+l)+\sqrt{-l^2(-2+l^2)}} - \frac{(1+l)\gamma}{4\sqrt{(1+l)^2\gamma^2}}$, when $\theta = 0$, $\beta_1 < \beta_3$; so when $\theta > 0$, $\beta_1 \leq \beta_1(\theta = 0) < \beta_3(\theta = 0) \leq \beta_3$;

If $\alpha < \alpha_0$, when $\theta = 0$, $\beta_1 < \beta_3$; when $\theta = \theta_4$, $\beta_1 = \beta_3$; so when $0 < \theta \leq \theta_4$, $\beta_1 \geq \beta_1(\theta = \theta_4) = \beta_3(\theta = \theta_4) > \beta_3$; when $\theta_4 < \theta < \gamma$, $\beta_1 \leq \beta_1(\theta = \theta_4) = \beta_3(\theta = \theta_4) \leq \beta_3$.

Proof of Proposition 9. The equilibrium solutions can be due to backward induction. The equilibrium strategies can be derived by a process similar to Proposition 7 so we omit it. Note that θ_5 is the first root of $\pi_m^{AY1*} - \pi_m^{AN1*} = 0$; θ_6 is the first root of $\pi_p^{WY1*} - \pi_p^{WN1*} = 0$; β_4 is the first root of $\pi_m^{AY1*} - \pi_m^{WY1*} = 0$; β_5 is the first root of $\pi_m^{WY1*} - \pi_m^{AN1*} = 0$; β_6 is the first root of $\pi_m^{WN1*} - \pi_m^{AN1*} = 0$.

Proof of Proposition 10.

$\theta_5 > \gamma$ and $\theta_6 > \frac{1-\alpha-\beta}{1-\beta} \gamma$, so the service is more likely to be adopted. The return volume is $(1-\delta)D_n^{k*}$, $k = WN1, WY1$. $D_n^{WY1*} > D_n^{WN1*}$. Therefore, the return volume is increased with pay-on-delivery service.

Denote that $B = \begin{cases} \beta_1, 0 < \theta \leq \frac{1-\alpha-\beta}{1-\beta} \gamma \\ \beta_2, \frac{1-\alpha-\beta}{1-\beta} \gamma < \theta \leq \gamma \\ \beta_3, \theta > \gamma \end{cases}$, $B' = \begin{cases} \beta_4, 0 < \theta \leq \theta_5 \\ \beta_5, \theta_5 < \theta \leq \theta_3 \\ \beta_6, \theta > \gamma \end{cases}$. $B' < B$ always satisfies.

Proof of Proposition 11.

$\pi_p^{WY2*} - \pi_p^{WN2*} = \frac{(\gamma-\eta\theta)}{16(1-\theta)(\gamma^2-8\gamma\eta+6\gamma\eta\theta+\eta^2\theta^2)^2} F''$, $F'' = l^2\gamma^3\theta - l^2\eta^3\theta^4 + \gamma\eta^2\theta(16(-1+\theta)^2 - 32l(-1+\theta)^2 + l^2(-4+\theta)(-4+3\theta)) - \gamma^2\eta(16(-1+\theta)^2 + 32l(-1+\theta)^2 + l^2(-4+\theta)(-4+3\theta)) > 0$. Therefore, $\pi_p^{WY2*} > \pi_p^{WN2*}$ when $\theta < \frac{\gamma}{\eta}$.

Denote $F_A'(l) = \pi_m^{AY2*} - \pi_m^{AN2*} = \frac{a\gamma^2+b\gamma+c}{B}$.

$$a = (1-\alpha-\beta)^2((1+l)^2(1-\beta) - (1+l(2-\beta)+\beta)\theta) > 0.$$

$$b = -(1-\alpha-\beta)\eta(l^2(2-(2-\beta)\beta) - l(2-\beta)\beta(2-\theta) + (2-(2-\beta)\beta)(1-\theta))\theta < 0.$$

$$c = (1-\beta)\eta^2\theta^2((1-l)^2 - \theta + l(2-\beta)\theta) > 0.$$

$$B = (4(-1+\beta) + (-2+\beta)^2\theta)((-1+\alpha+\beta)^2\gamma^2 - 2(-1+\alpha+\beta)\gamma\eta(-2+\theta) + \eta^2\theta^2) > 0.$$

Let $a\gamma^2 + b\gamma + c = 0$, we can obtain $\gamma = \frac{\eta\theta((-1+l)^2-(1+l(-2+\beta))\theta)}{(-1+\alpha+\beta)((1+l)^2(-1+\beta)-(-1+l(-2+\beta)+\beta)\theta)}$ and $\gamma =$

$$\frac{(1-\beta)\eta\theta}{1-\alpha-\beta}, \text{ where } \frac{\eta\theta((-1+l)^2-(1+l(-2+\beta))\theta)}{(-1+\alpha+\beta)((1+l)^2(-1+\beta)-(-1+l(-2+\beta)+\beta)\theta)} < \frac{(1-\beta)\eta\theta}{1-\alpha-\beta}.$$

Therefore, $F'_A(l) > 0$ when $\gamma < \frac{\eta\theta((-1+l)^2-(1+l(-2+\beta))\theta)}{(-1+\alpha+\beta)((1+l)^2(-1+\beta)-(-1+l(-2+\beta)+\beta)\theta)}$ or $\gamma > \frac{(1-\beta)\eta\theta}{1-\alpha-\beta}$. $\gamma < \frac{\eta\theta((-1+l)^2-(1+l(-2+\beta))\theta)}{(-1+\alpha+\beta)((1+l)^2(-1+\beta)-(-1+l(-2+\beta)+\beta)\theta)}$ can never be satisfied under the condition that the market is partially covered. Therefore, $\pi_m^{AY2*} > \pi_m^{AN2*}$ when $\gamma > \frac{(1-\beta)\eta\theta}{1-\alpha-\beta}$, i.e., $\theta < \frac{1-\alpha-\beta}{(1-\beta)\eta}\gamma$.

Proof of Proposition 12. The proof of Proposition 12 is similar to that of Proposition 7, so we omit it. Note that β_7 is the first root of $\pi_m^{AY2*} - \pi_m^{WY2*} = 0$; β_8 is the first root of $\pi_m^{WY2*} - \pi_m^{AN2*} = 0$; β_9 is the first root of $\pi_m^{WN2*} - \pi_m^{AN2*} = 0$.